Which explanations can we trust? A multi-method faithfulness, cross-method agreement, and anatomical-validity audit of a frozen pretrained SFCN on pediatric brain MRI

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Abstract

Explanation methods are routinely used to justify clinical claims from 3D brain-MRI classifiers, yet they are rarely validated before being believed. We audit nine attribution methods on a frozen pretrained Simple Fully Convolutional Network (SFCN), a brain-age model, using an age-group proxy task on 96 children from the public Calgary Preschool control cohort. The study makes no claim about prenatal alcohol exposure; it asks which explanations are faithful to the model, agree with one another, and yield region-level claims that survive multiple-comparison correction. Occlusion-based ReX was the most faithful (deletion AUC 0.14 vs. 0.53–0.78 for the rest) and early-block Grad-CAM the least. Eight of nine methods passed a randomized-weight sanity check; Integrated Gradients did not (Spearman $r = 0.73$ between trained and random models), indicating input-driven rather than model-driven maps. Methods agreed only weakly (mean pairwise Spearman 0.36), and the two Grad-CAM variants were anti-correlated. A corrected region-masking test identified 15 cortical regions under an exploratory zero-fill, but none under the pre-specified mean-intensity fill, so this positive result is post-hoc and fill-dependent. Method choice materially changes the regions one would report, and occlusion-based faithfulness and randomized-weight sanity checks should be routine. The contribution is a reusable validation protocol, not a diagnostic classifier.

Keywords: explainable AI, saliency maps, faithfulness, sanity checks, pediatric MRI, brain age, SFCN, prenatal alcohol exposure.

1 Introduction

Deep classifiers for 3D brain MRI are increasingly accompanied by saliency or class-activation maps that purport to show *where* the network looks. When those maps are read anatomically ("the model attends to the corpus callosum"), they migrate quickly into clinical narrative. But attribution methods are known to disagree with each other and, in some cases, to produce the same map whether or not the model has learned anything (Adebayo et al., 2018). Before an explanation is trusted enough to support a claim about a brain region, three questions should be answered: is it *faithful* (do the voxels it highlights actually drive the model?), is it *consistent* (do independent methods agree on which regions matter?), and is it *anatomically valid* under a corrected statistical

test? We use "consistency" in the specific sense of cross-method agreement; we do not test map stability under input perturbation or random seeds, which we flag as a limitation (Section 6).

Das et al. (2024) trained a frozen pretrained SFCN to detect prenatal alcohol exposure (PAE) from pediatric brain MRI and reported explanations as a *preliminary* step: a single Grad-CAM method, with only one named region (white-matter volume) formally tested ($t = -0.73$, $p = 0.47$). The PAE scans are not public. This work does not attempt to reproduce their clinical result. Instead, it takes the identical frozen SFCN backbone and asks the three validation questions above on a task the public data can actually support: an age-group proxy, which is appropriate because the frozen network is a brain-age model and age is the signal its features encode.

Our contributions are: (i) a pre-specified, multi-method validation protocol for 3D-MRI attribution combining faithfulness, randomized-weight sanity checks, resolution, cross-method agreement, and a corrected anatomical region test; (ii) an empirical verdict over nine methods on a real neuroimaging backbone; and (iii) a public, reproducible pipeline. The headline finding is that *method choice is not a detail*: faithfulness spans a factor of five across methods, one widely used method (Integrated Gradients) fails the sanity check, and methods agree only weakly on which regions matter.

The paper follows the three questions as a spine. We answer *faithful?* with deletion and insertion AUC (Section 4.2), *model-driven rather than input-driven?* with a randomized-weight sanity check (Section 4.3), *consistent?* with cross-method agreement (Section 4.5), and *anatomically valid?* with a multiple-comparison-corrected region-masking test (Section 4.6); resolution (Section 4.4) measures whether a map is fine enough to name a structure at all.

2 Related work

2.1 Attribution methods and why they must be validated

Attribution methods assign each input voxel a score for how much it drove a prediction, but they compute that score in very different ways, and the differences matter for what a map can be trusted to show. *Gradient* methods (Saliency, Integrated Gradients (Sundararajan et al., 2017), SmoothGrad (Smilkov et al., 2017), VarGrad (Adebayo et al., 2018)) read the sensitivity of the output to each voxel and so produce voxel-resolution but often noisy maps. *Class-activation* methods (Grad-CAM (Selvaraju et al., 2017)) weight the network's last convolutional feature maps, yielding smooth, coarse localizations at the resolution of that feature grid. *Relevance-propagation* methods (LRP (Bach et al., 2015)) redistribute the output backward through the network under conservation rules. *Perturbation/occlusion* methods (RISE (Petsiuk et al., 2018); here, ReX (Chockler et al., 2024)) instead mask parts of the input and watch the prediction move, so they test the model directly rather than reading its internals. Because these mechanisms differ, the same scan can yield maps that disagree on which region matters, which is exactly what makes a validation step necessary rather than optional.

A body of work shows that a visually convincing map need not reflect anything the model learned. Adebayo et al. (2018) found that several popular methods are largely invariant to randomizing the model's weights, so their output cannot be explaining learned structure; our randomized-weight sanity check follows that logic. Kindermans et al. (2019) showed saliency can change under input shifts that leave the model unchanged, and Ghorbani et al. (2019) that interpretations are fragile to imperceptible perturbations. To move past appearance, two functional criteria are now standard: occlusion-based faithfulness, measured as the prediction drop when the highlighted voxels

are removed (deletion/insertion (Petsiuk et al., 2018)) or as retraining-based importance (ROAR (Hooker et al., 2019)). Critically, this validation literature is almost entirely 2D natural-image classification. The handful of medical-imaging studies that test attribution robustness (Eitel and Ritter, 2019; Arun et al., 2021) report that saliency maps are often unreliable for localization, yet 3D neuroimaging attributions are still routinely read anatomically without any such check. That gap, a full multi-criterion validation of attribution on a real 3D brain-MRI backbone, is what this paper addresses.

2.2 Relation to Das et al. (2024)

This is independent research, carried out by an author not affiliated with the original study, that builds on and extends the explainability step that Das et al. (2024) explicitly flagged as preliminary. The original contribution, the frozen-SFCN transfer-learning approach to prenatal alcohol exposure and its clinical framing, is theirs; our contribution is the validation protocol applied to the same backbone on public data. It is not an audit of their clinical conclusion, which we cannot access, and every use we make of their design is cited as such. We identify six open questions about the original analysis (coarse explanation grid, single-region testing, site/recruitment confound, age confound, child-level leakage from repeated scans, and a scan-count-versus-child-count clarification), listed here and then stated and revisited one by one in Section 5.3 to the extent the public data allow. The complete preregistered protocol (OSF `jycnr`) accompanies this work in the project repository.

3 Methods

3.1 Data and cohort

We use the public Calgary Preschool control cohort (Reynolds et al., 2020) (OSF fork `5kz2p`, CC-BY 4.0), restricted to the 96 children who carry a demographic label and a usable T1 scan, one scan per child. The public release is longitudinal and lists multiple visits per child, so its total T1 count exceeds the number of children; we take one scan per child, which removes child-level leakage. Cohort demographics are in Table 1. The proxy task is age dichotomized at the cohort median (3.64 y): 48 older (label 1), 48 younger (label 0), a perfectly balanced binary problem (Fig. 9a).

Table 1: Cohort characteristics ($n = 96$ children, one scan each).

Characteristic	Value
N children	96
Age, years, mean (SD)	3.84 (0.83)
Age, years, range	2.58 to 6.90
Male, n (%)	50 (52.1)
Female, n (%)	46 (47.9)

3.2 Preprocessing

Each volume underwent ANTs (Avants et al., 2011) N4 bias correction (Tustison et al., 2010), antspynet deep-learning brain extraction (Tustison et al., 2021), affine registration to 1 mm MNI152,

and cropping to $160 \times 192 \times 160$, the input space in which the SFCN was pretrained. All 96 volumes passed visual QC (Fig. 9b). This is a deliberate design decision and differs from Das et al. (2024), who registered to a pediatric atlas and used FSL-BET + SynthStrip. Because our backbone is frozen and never fine-tuned, its features are only valid on inputs drawn from the adult UK Biobank/MNI152 distribution it was trained on; registering the children into a pediatric template would move the inputs off that distribution and invalidate the very features we are auditing. Keeping the model's native space is therefore the methodologically consistent choice for a frozen backbone. Its one cost is that an adult atlas is then applied to preschool brains through affine-only registration, so region-level anatomical labels are approximate; we bound that cost in Section 6.

3.3 Model

We use the original SFCN (Peng et al., 2021), a lightweight fully convolutional brain-age network, with its six convolutional blocks frozen. The frozen backbone uses the UK Biobank brain-age weights released by Peng et al. (2021) in their official repository (Peng, 2021) (checkpoint `brain_age/run_20190719_00_epoch_best_mae.p`); we did not retrain the backbone. A 512-unit head with a 2-logit softmax replaces the paper's PAE head and is trained for the age-group proxy. All attributions are computed with respect to this frozen-backbone, trained-head model, using the head trained on the full 96-child cohort; the cross-validation in Section 4.1 is a separate accuracy estimate and does not feed the attribution model.

3.4 Explanation methods

We compare nine methods (Table 2), spanning gradient, class-activation, relevance-propagation, and occlusion families. M8 (Guided Backprop (Springenberg et al., 2015)) is included as a negative control; M1 (final-block Grad-CAM) is the method used by Das et al. (2024). Captum (Kokhlikyan et al., 2020) provides M1–M6 and M8; zennit (Anders et al., 2021) provides M7; M9 is ReX (Chockler et al., 2024), a 3D occlusion-responsibility search on the $5 \times 6 \times 5$ feature grid. All attributions are computed toward each scan's true proxy-label class, on all 96 children. Integrated Gradients uses a zero-image baseline and 32 steps; SmoothGrad and VarGrad use 20 noised samples ($\sigma = 0.1$); Grad-CAM targets the final (M1) and an earlier (M2) convolutional block.

Table 2: The nine explanation methods.

ID	Method	ID	Method
M1	Grad-CAM, final block (paper's)	M6	VarGrad
M2	Grad-CAM, earlier block	M7	LRP (EpsilonPlusFlat)
M3	Saliency	M8	Guided Backprop (control)
M4	Integrated Gradients (32 steps)	M9	ReX (3D occlusion)
M5	SmoothGrad (20 samples)		

3.5 Metrics and statistics

Each metric below answers one of the three framing questions; two supporting metrics (resolution, and the classifier/probe checks) tell us whether the map and the proxy are even interpretable.

Table 3 is the map from metric to question to pass criterion; the paragraphs that follow give the detail.

Table 3: What each metric measures, the question it answers, and what counts as a pass.

Metric	Question answered	Pass / reading
Faithfulness	Do the highlighted voxels drive the output?	Low deletion AUC, high insertion AUC
Sanity	Is the map model-driven, not input-driven?	$\|r\| < 0.3$ between trained and random model
Agreement	Do independent methods agree on regions?	High pairwise region-level r
Region test	Which regions is the *model* sensitive to, under correction?	Survives BH-FDR $q = 0.05$ vs. random-region null
Resolution	Is the map fine enough to name a structure?	Supporting: top-5% mass, blob count
Classifier / probe	Is the proxy signal real (not noise)?	Supporting: above-chance CV, permutation p

Faithfulness (question 1). Deletion and insertion AUC (Petsiuk et al., 2018) (lower deletion AUC is better). This is a partial-deletion curve, not a full 0–100% sweep: we remove the top-ranked voxels at four fractions (5, 10, 20, 30%) with a zero fill and integrate the predicted-class probability over that range. We cap at 30% because the discriminative signal is concentrated in the top voxels and deleting a third of the brain already drives most methods to chance; the AUC should be read as "area over 5–30% deletion," not the conventional full-curve value. If a map is faithful, deleting what it ranks highest should collapse the prediction fastest.

Sanity (question 2). Spearman correlation of a method's map between the trained model and a fully reinitialized model (Adebayo et al., 2018); low correlation is good. The reinitialization is complete, not head-only: we deep-copy the model and call `reset_parameters()` on every module that defines it, so *all six convolutional backbone blocks and the classifier head* are reset from a fixed seed. This matters for interpreting the result: a method that stays correlated here is unchanged even when the entire network, including the frozen feature extractor, is random, so its map cannot be reflecting learned structure. We use the standard complete-randomization variant; the layer-by-layer cascading variant of Adebayo et al. (2018) was not run (Section 6). Maps are compared as the per-voxel absolute attribution, with a single Spearman correlation taken over all voxels of the two maps; the random model is a single reinitialization from a fixed seed, so the check uses one random draw rather than an average over seeds. A method is scored PASS when $|r| < 0.3$; this band is a pre-specified heuristic, not a published constant, and is drawn as the dotted line in Fig. 3.

Agreement (question 3, part A). Pairwise Spearman correlation of region-level scores: if the methods are measuring the same underlying signal, their regional rankings should line up.

Region test (question 3, part B). In plain terms, this test asks which brain regions the *model* actually depends on, and it does so *without reference to any saliency map*: for each Harvard–Oxford cortical region (Desikan et al., 2006; Makris et al., 2006) we blank the region's voxels and record the drop in predicted-class probability. A large drop means the model relied on that region. This is why it is a masking test on model output, not an attribution test; it validates the model's regional dependence directly, so it can confirm or refute what the maps claim. Significance is against a null of 200 random same-size regions, with Benjamini–Hochberg correction at $q = 0.05$ (Benjamini and

Hochberg, 1995). We run two fills, mean-intensity and zero; because every effect reported below is fill-dependent (only the zero fill yields any significant region), the results and Table 6 report the zero-fill effect, with the mean-fill outcome given alongside.

Resolution (supporting). Fraction of attribution mass in the top 5% of voxels, and blob count, how localized a map is, hence whether it is fine enough to name a structure.

Uncertainty and significance. 95% bootstrap CIs (10,000 resamples, resampled by child) on every metric. The classifier check is a repeated (10×) stratified 5-fold CV grouped by child, plus a 1,000-permutation test against chance, establishing that the proxy signal is real before any map is interpreted. All tests were logged with raw and corrected p-values; tests run after seeing the data are flagged exploratory in the pre-specified plan.

4 Results

4.1 Proxy classifier (a sanity check, not a result)

The head need only learn the proxy well enough to make attributions meaningful; its accuracy is not a scientific claim. Cross-validated accuracy was 0.643 ± 0.120 over 50 fits (sensitivity 0.632, specificity 0.668), and a linear probe on the frozen features reached 0.635 (permutation $p = 0.01$). The proxy is decodable above chance but only modestly (Fig. 1).

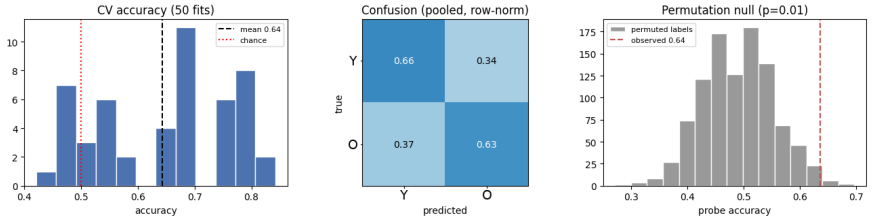


Figure 1: Proxy-task classifier performance (age group from frozen SFCN features).

4.2 Faithfulness

Addressing question 1 (do the highlighted voxels drive the output?), ReX (M9) is clearly the most faithful method: deletion AUC 0.141 ([0.119, 0.165]) against 0.53–0.78 for the rest. This advantage is partly by construction and we state it plainly: ReX searches for regions whose removal changes the prediction, and deletion AUC scores exactly that property, so the two are not independent. The insertion AUC (0.755), a criterion ReX does not optimize, points the same way and is the less circular evidence that its voxels drive the output. Only the extremes are individually separable: ReX at the top, and LRP (M7) and early-block Grad-CAM (M2) at the bottom. The five mid-ranked methods (Integrated Gradients, Guided Backprop, SmoothGrad, Saliency, VarGrad) have heavily overlapping bootstrap CIs and should be read as a tie, not a rank order. The paper's method (M1) is middling, and early-block Grad-CAM (M2) is the worst of the nine (Table 4, Fig. 2).

Table 4: Faithfulness. Deletion AUC (lower is better) with 95% bootstrap CI; insertion AUC.

Method	Deletion AUC	95% CI	Insertion AUC
M9 ReX	**0.141**	[0.119, 0.165]	0.755
M4 Integrated Gradients	0.528	[0.475, 0.584]	0.512
M8 Guided Backprop	0.556	[0.489, 0.624]	0.742
M5 SmoothGrad	0.556	[0.502, 0.609]	0.537
M3 Saliency	0.568	[0.518, 0.619]	0.490
M6 VarGrad	0.580	[0.531, 0.628]	0.501
M1 Grad-CAM (final)	0.644	[0.601, 0.687]	0.501
M7 LRP	0.674	[0.627, 0.720]	0.598
M2 Grad-CAM (early)	**0.777**	[0.730, 0.822]	0.488

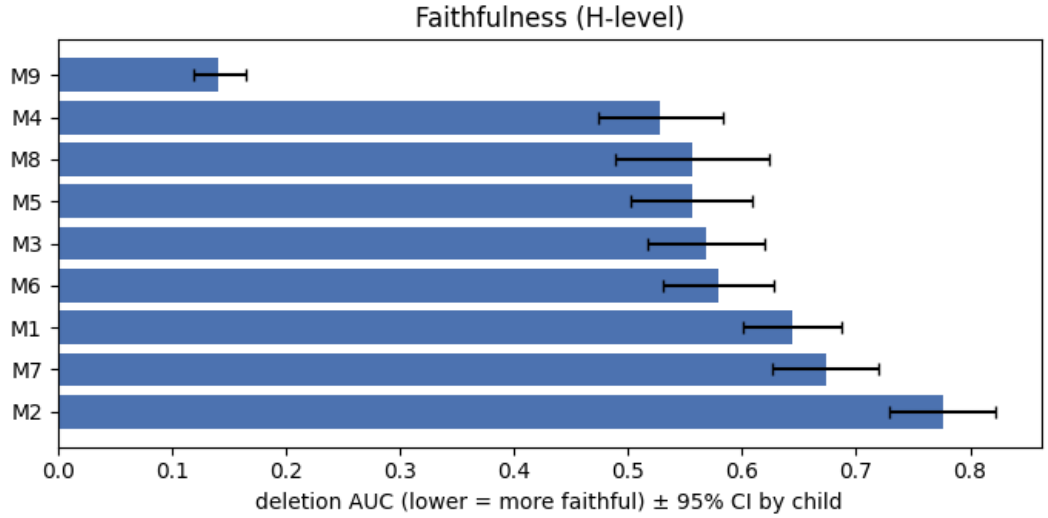


Figure 2: Deletion and insertion AUC across the nine methods.

4.3 Randomized-weight sanity check

Addressing question 2 (is the map model-driven rather than input-driven?), eight methods pass (low trained-vs-random correlation). Integrated Gradients does not ($r = 0.73$): its map is almost unchanged when every layer, backbone and head, is reinitialized, so on this network it reflects image structure rather than learned content (Table 5, Fig. 3). Because the randomization is complete rather than head-only, this correlation cannot be an artifact of an unchanged frozen backbone: the backbone is random too. Part of this is mechanical and worth stating: with the zero baseline we use, Integrated Gradients reduces to the input-times-gradient form, so a dependence on input intensity is built into the method and survives weight randomization. The practical caution is nonetheless real, because IG is widely used in neuroimaging (Eitel and Ritter, 2019) and its maps look convincing.

Guided Backprop (M8), our negative control, passes here ($r = 0.168$), which is at first surprising: Adebayo et al. (2018) report it as largely invariant to model randomization and therefore a method that typically *fails* this check. The reconciliation cuts two ways, and the second way is a caveat

on our own test. The verdict depends on architecture and implementation: Guided Backprop's insensitivity was shown on 2D ImageNet CNNs with ReLU backprop overrides, and on a frozen six-block 3D SFCN with a small trained head the trained-vs-random map is less correlated. But a control passing where the literature expects it to fail also means our single-draw, complete-randomization check is less sensitive here than the cascading variant would be, so the PASS column should be read as a coarse filter, not a certificate. Passing it is in any case necessary, not sufficient: M8 is still a poor explanation on the metric that matters, with a near-chance deletion AUC (0.556).

Table 5: Randomized-weight sanity check. Spearman correlation, trained vs. random model (lower is better).

Method	Spearman (trained vs. random)	Verdict
M2 Grad-CAM (early)	−0.242	PASS (model-dependent)
M9 ReX	0.019	PASS
M1 Grad-CAM (final)	0.121	PASS
M3 Saliency	0.141	PASS
M6 VarGrad	0.153	PASS
M8 Guided Backprop	0.168	PASS
M5 SmoothGrad	0.172	PASS
M7 LRP	0.255	PASS
M4 Integrated Gradients	**0.728**	**FAIL (input-driven)**

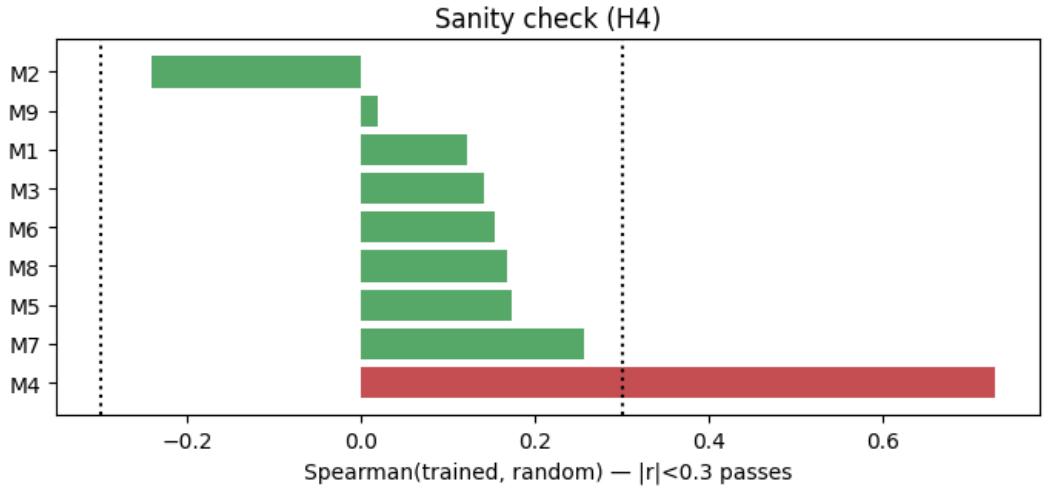


Figure 3: Sanity check: similarity between trained-model and random-model attributions.

4.4 Resolution

The two methods that operate on the $5 \times 6 \times 5$ feature grid, final-block Grad-CAM (M1) and ReX (M9), produce only about three large blobs each; in Fig. 4 their blob counts sit near zero against a scale dominated by the voxel-level methods, which fragment into thousands of small blobs (SmoothGrad ≈19,000, Saliency ≈44,000). The near-zero bars for M1 and M9 are therefore expected, not a plotting error: coarse grid-level maps cannot have many blobs. This directly bears

on the first open question about the original analysis, namely that the explanation grid is coarser than many named structures.

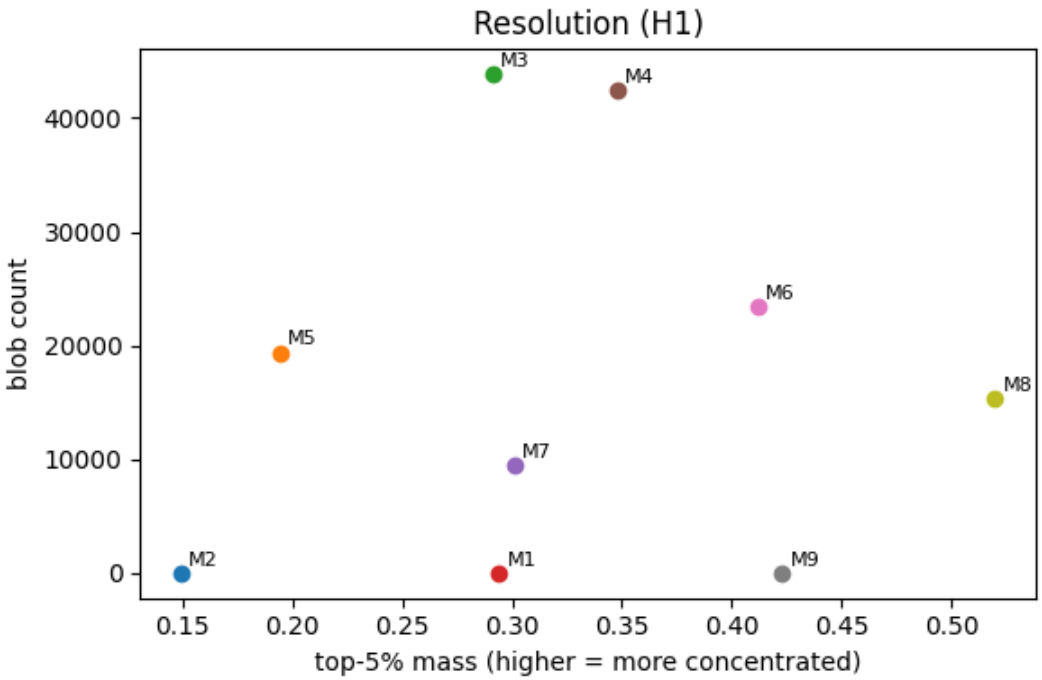


Figure 4: Resolution: top-5% attribution mass and blob count per method. Grid-level methods (M1 Grad-CAM final, M9 ReX) show ≈3 blobs, appearing near zero on the voxel-level scale; this is inherent to the $5 \times 6 \times 5$ grid, not an artifact.

4.5 Cross-method agreement

Mean pairwise region-level Spearman correlation is only 0.36 (Fig. 5). The gradient family (M3/M5/M6) agrees strongly among itself, with pairwise r reaching 0.96 at the top of its range (reported as the maximum, not a typical value), but the two Grad-CAM variants are anti-correlated ($r = -0.61$), and ReX and LRP sit largely on their own. Which regions one reports depends heavily on the method chosen.

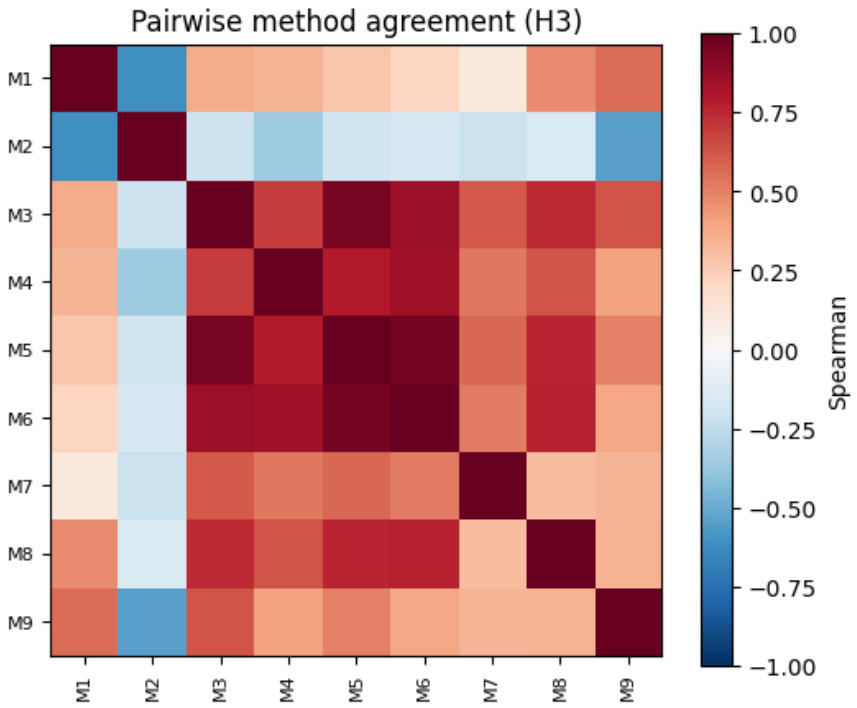


Figure 5: Pairwise cross-method agreement (region-level Spearman correlation).

4.6 Anatomical region test

Under an exploratory zero-fill (a post-hoc deviation from the pre-specified mean-intensity fill; see Section 5.2), fifteen cortical regions survive Benjamini–Hochberg correction at $q = 0.05$ (Table 6, Fig. 6); the Lingual Gyrus carries the largest effect. The pre-specified result is the opposite: under the mean-intensity fill no region survives correction (smallest raw $p = 0.95$). That $p = 0.95$ is expected, not anomalous: the test is two-sided on the magnitude of the probability drop, and under the mean-intensity fill every region's observed drop is near zero and smaller than almost all same-size random-region drops, so mean-intensity replacement is too gentle a perturbation to expose regional dependence, not evidence that the regions are unimportant. The zero fill has the opposite caveat: blanking a region to zero is itself out of distribution, so part of its larger effect may reflect how much signal is removed rather than the region's specific importance. The same-size random-region null absorbs the gross "how much brain" component, but not the out-of-distribution character of a zero patch. The surviving regions should therefore be read as significant for zero-region deletion specifically, an exploratory finding, not a fill-robust anatomical claim. The informative column is the effect size, not the FDR p: with a null of 200 random regions the smallest attainable corrected p is a floor (hence the repeated 0.017), so the p column separates "survives" from "does not" but does not rank the regions. The full per-region log (both fills, raw and corrected p) is released as `results/region_masking_all_pvalues.csv`.

Table 6: The 15 cortical regions surviving BH-FDR $q = 0.05$ under zero-fill masking. Effect is the zero-fill masking effect size (probability drop); n is region voxel count. The FDR p is at the floor set by the 200-region null (see text), so effect size, not p, orders the regions. No region survives under the mean-intensity fill.

Region	Effect	n voxels	FDR p
Lingual Gyrus	0.257	26,880	0.017
Temporal Fusiform Cortex, posterior div.	0.111	12,680	0.017
Insular Cortex	0.081	21,449	0.017
Parahippocampal Gyrus, posterior div.	0.069	5,698	0.017
Inferior Temporal Gyrus, temporooccipital	0.063	11,973	0.017
Cingulate Gyrus, anterior div.	0.059	20,844	0.017
Parahippocampal Gyrus, anterior div.	0.057	9,956	0.017
Cingulate Gyrus, posterior div.	0.057	19,228	0.017
Occipital Fusiform Gyrus	0.053	14,538	0.017
Central Opercular Cortex	0.044	14,953	0.017
Parietal Operculum Cortex	0.043	8,739	0.017
Subcallosal Cortex	0.039	9,021	0.017
Frontal Operculum Cortex	0.035	5,313	0.017
Supramarginal Gyrus, posterior div.	0.029	18,720	0.032
Planum Temporale	0.025	8,032	0.017

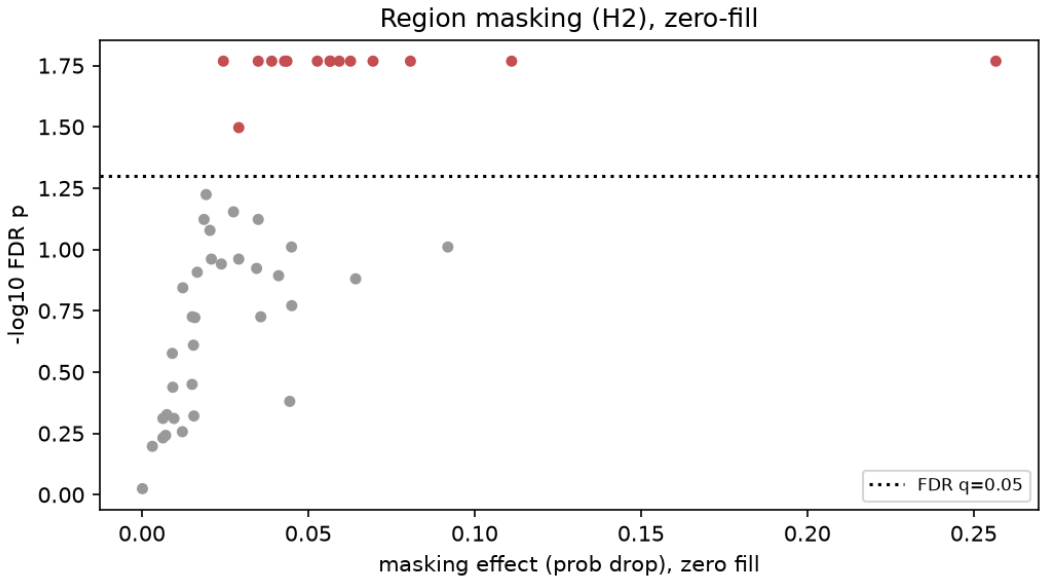


Figure 6: Region masking test (zero fill) against a random-region null: zero-fill masking effect size vs. $-\log_{10}$ FDR p. Red points survive BH-FDR $q = 0.05$. The x-axis matches the effect column of Table 6.

4.7 Layer probes and qualitative example

Linear-probe accuracy was 0.624 from block 2 and 0.635 from block 6 (Fig. 7): depth adds little for this task, and the proxy is almost as decodable from an early block as from the final one. A representative scan across all nine methods (Fig. 8) illustrates the resolution and agreement differences quantified above.

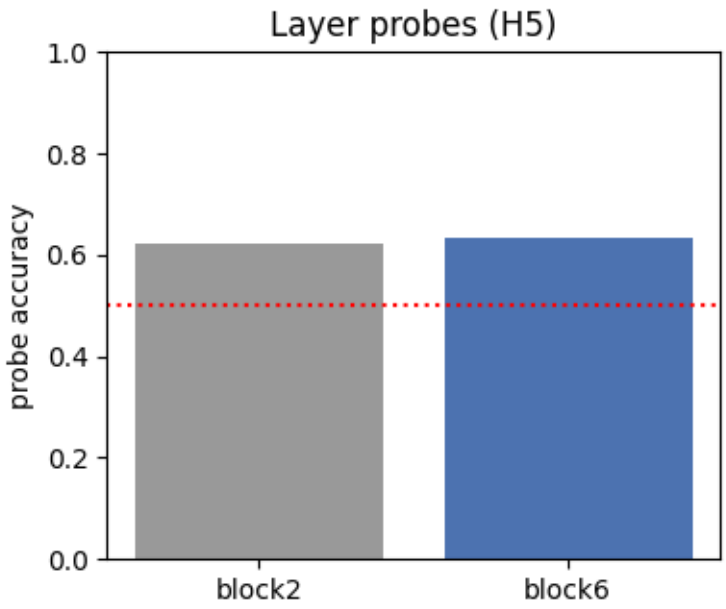


Figure 7: Linear-probe accuracy by feature block (block 2 vs. block 6).

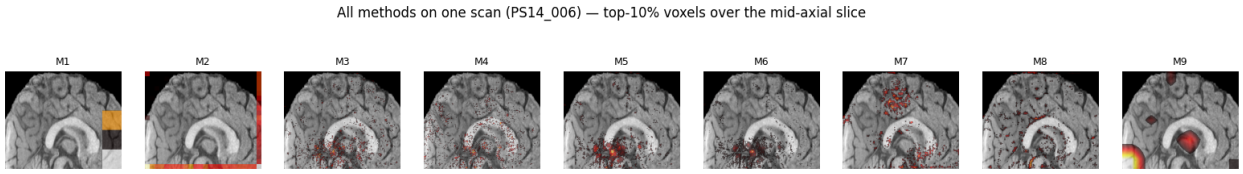


Figure 8: Qualitative attribution maps for one representative scan across all nine methods, on matched slices. The resolution and agreement differences quantified in Figs. 4 and 5 are visible here.

5 Discussion

5.1 Pre-specified hypotheses

Results are reported against the six hypotheses exactly as registered (OSF `jycnr`).

H1 (Grad-CAM produces smoother, coarser maps than Integrated Gradients and SmoothGrad): *supported on blob count, mixed on mass.* Both Grad-CAM variants resolve to only ≈3 connected blobs, versus tens of thousands for Integrated Gradients and SmoothGrad (Fig. 4); this blob-count gap is the unambiguous evidence for "coarser." Top-5% attribution mass does not discriminate cleanly: early-block Grad-CAM (M2) carries the lowest mass, but final-block Grad-CAM (M1, ≈0.30) is *more* concentrated than SmoothGrad (≈0.19), so we rest the claim on blob count rather than mass.

H2 (large anatomical regions are more likely to survive correction than small named structures such as the pons and corpus callosum): *not supported*; under the pre-specified mean-intensity fill no region survived BH-FDR ($q = 0.05$), so the large-versus-small contrast could not be tested; a zero-fill sensitivity check yielded 15 surviving regions, confirming the region test is fill-dependent.

H3 (Integrated Gradients and SmoothGrad correlate more with each other than either does with final-block Grad-CAM): *supported descriptively*; mean pairwise region-level Spearman is 0.36, and the gradient family (including IG and SmoothGrad) agrees internally (up to $r = 0.96$) while Grad-CAM sits apart. This is reported descriptively; no formal test of the difference between dependent correlations was pre-specified or run.

H4 (for at least one method, randomized-weight maps differ from trained-weight maps): *supported*; most methods change substantially under weight randomization; Integrated Gradients is the lone weak case, retaining the highest trained-versus-random similarity ($r = 0.73$).

H5 (a block-6 linear probe beats a block-2 probe): *not supported*; 0.635 vs. 0.624 is a 0.011 gap, far inside the cross-validation standard deviation (± 0.12), so depth confers no measurable advantage for this proxy and the two probes should be read as tied.

H6 (age-group accuracy exceeds chance under a child-level label-permutation test): *supported*; the child-level label-permutation test was run on the frozen-feature linear probe (accuracy 0.635, permutation $p = 0.01$, 1,000 permutations), and the repeated-CV head reaches 0.643 ± 0.120. The proxy signal is usable but modest; 0.635 is the probe accuracy that the permutation test refers to, not the CV-head accuracy.

5.2 Deviations and exploratory analyses

The following are not part of the registered hypotheses and are reported as exploratory: (i) the count of regions surviving correction and its dependence on the masking fill, an elaboration of H2; (ii) whether age is independently decodable from the frozen features as a possible confound (Ridge $R^2 = 0.11$); and (iii) fine-tuning the SFCN, which was pre-specified as out of scope and not run. Choices made after inspecting results are flagged as exploratory here and in the project decision log.

5.3 The six open questions about the original analysis

1. **Coarse explanation grid**, *confirmed*: Grad-CAM resolves to a few blobs on the $5 \times 6 \times 5$ grid, coarser than named structures.
2. **Single region tested**, *a corrected multi-region test is feasible*: it finds 15 regions under zero-fill masking (none under mean-fill). This is a different test from the original white-matter-volume t-test, not a like-for-like correction of it, so we draw no comparison of effect direction or magnitude with Das et al. (2024); the point is only that a multi-region, FDR-controlled test can be run where a single region was.
3. **Site/recruitment confound**, *not testable*: PAE scans are not public.
4. **Age confound**, *partially testable*: age is weakly decodable from the frozen features ($R^2 = 0.11$); a plausible confound worth controlling.
5. **Child-level leakage**, *addressed by design*: one scan per child and child-grouped CV.
6. **Scan count vs. child count**, *a clarification of our own n, not a criticism of the original study*: the public release is longitudinal and lists multiple visits per child. Our reading of that release is that its total T1 count therefore exceeds the number of unique children (by our count on the order of 279 scans against 96 children), though this scan total is our inference from the

public visit structure, not a figure stated by Das et al. (2024). We include it only to explain why our analysis is child-level ($n = 96$, one scan each); we make no claim that any count in the original work is incorrect.

5.4 What this means in practice

Method choice materially changes the reported "important regions." Three practices follow, one per framing question: for faithfulness, report occlusion-based deletion/insertion rather than trusting a map's appearance; for model-dependence, run a randomized-weight sanity check and discard methods that fail it (here, Integrated Gradients); and for anatomical validity, never report a single method's regions without a corrected statistical test.

6 Limitations

$n = 96$; a single proxy task; one cortical atlas; and no PAE data. One caution follows from the native-space design justified in the Methods: the adult Harvard–Oxford atlas reaches the preschool brains through affine-only registration, so for children aged 2.6–6.9 y the region identities in Table 6 are approximate anatomical labels, and the anatomical-validity claim holds at the resolution of the masking test rather than of individual gyri. Age-appropriate pediatric templates and nonlinear registration would sharpen the labels (Fonov et al., 2011) but would invalidate the frozen features, so the trade-off is deliberate, not an oversight. The backbone weights are the public UK Biobank brain-age release of Peng et al. (2021) (Peng, 2021); we report the checkpoint used rather than relying on an unverified source. The proxy head is weak (≈64% CV accuracy), and faithfulness and sanity verdicts are properties of the model being explained: a stronger classifier could rank the methods differently, so the rankings here should be read as holding for this modest-accuracy model, not as universal. At $n = 96$ the study is powered for medium-to-large region effects; small effects may be inconclusive (H3 and H4 do not depend on effect size). Three gaps bound the interpretation. First, we measured cross-method *agreement* but not true map *stability*: we did not quantify how a single method's map varies under small input perturbations or across random seeds, so claims about reproducibility within a method are out of scope. Second, our sanity check uses complete model randomization; we did not perform the layer-by-layer cascading randomization of Adebayo et al. (2018), which would localize where in the backbone a method's dependence breaks down. Third, we report the method ranking against the observed spread of deletion AUCs rather than against an explicit random-attribution floor; a random-map baseline (expected deletion AUC near 0.5) would further anchor the weak methods but is not required for the comparative ranking reported here. This work validates an explanation protocol on an age-group proxy in public controls; it does not speak to alcohol exposure, FASD, or the original paper's clinical regions, which require the private PAE data.

7 Conclusion

On a frozen pretrained SFCN, nine explanation methods gave sharply different accounts of what the model uses. Occlusion-based ReX was the most faithful; Integrated Gradients failed a basic sanity check; methods agreed only weakly; and a corrected masking test isolated 15 cortical regions under an exploratory zero-fill, but none under the pre-specified mean-intensity fill, so that positive

result is post-hoc. The practical message for neuroimaging XAI is that attribution maps must be validated for faithfulness, model-dependence, and anatomical significance, before any of them is read as evidence about the brain.

Acknowledgments

This work builds on Das et al. (2024), whose study motivated the present validation and whose frozen-SFCN transfer-learning design we reuse on public data; we thank those authors for clearly flagging their explainability step as preliminary, which is what this paper extends. We also thank Reynolds et al. (2020) and the Calgary Preschool MRI study for releasing the control cohort under CC-BY 4.0, and Peng et al. (2021) for the public SFCN brain-age weights. The author declares no affiliation with the original study and no competing interests.

Reproducibility

Data: public Calgary Preschool controls (OSF fork `5kz2p`, CC-BY 4.0). Code, the pre-specified plan, all figures, and the full per-region p-value log are in the project repository: `https://github.com/HoussemDs/brain-xai-faithfulness-audit`. The full pipeline ran on Kaggle (GPU T4 $\times 2$, ≈ 5.8 h).

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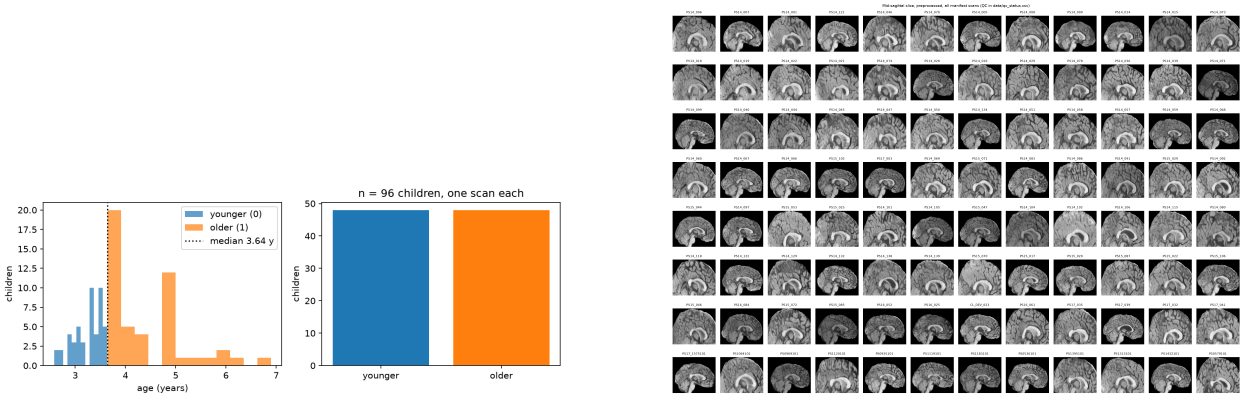
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A Supplementary preprocessing figures



(a) Age-group split at the cohort median.

(b) QC contact sheet (all 96 scans).

Figure 9: Cohort and quality control.